

Table 1: Performance comparison on seen and unseen datasets. Best results in **bold**, second-best underlined. “No KAN” = KAN-AINet without KAN module. Bottom block shows average scores across all unseen datasets.

Dataset	Method	mDice $\uparrow$	mIoU $\uparrow$	S_{α} $\uparrow$	F_{β}^w $\uparrow$	MAE $\downarrow$	HD ₉₅ $\downarrow$	ASD $\downarrow$
Kvasir-Sessile (seen)	nnU-Net	0.649	0.540	0.775	0.834	0.072	122.34	48.03
	ESPNet	0.887	0.818	0.920	0.916	0.020	36.76	<u>8.78</u>
	No KAN	0.856	0.785	0.909	0.911	0.024	54.10	20.50
	KAN-IMM only	0.890	0.824	0.925	0.920	0.018	<u>35.20</u>	11.40
	KAN-BAM only	0.876	0.807	0.919	0.919	0.021	43.20	13.00
	KAN-AINet (Ours)	<u>0.887</u>	<u>0.818</u>	<u>0.922</u>	<u>0.918</u>	<u>0.019</u>	31.20	7.40
CVC-ColonDB (unseen)	nnU-Net	0.608	0.505	0.763	0.777	0.055	129.77	55.57
	ESPNet	0.816	0.738	0.888	0.859	0.032	47.55	21.70
	No KAN	0.731	0.657	0.830	0.759	0.044	120.50	89.20
	KAN-IMM only	0.775	0.704	0.856	0.799	0.040	82.90	56.70
	KAN-BAM only	0.747	0.675	0.846	0.782	0.041	105.00	77.80
	KAN-AINet (Ours)	<u>0.784</u>	<u>0.711</u>	<u>0.862</u>	<u>0.811</u>	<u>0.039</u>	<u>78.30</u>	<u>52.00</u>
ETIS-Larib-PolypDB (unseen)	nnU-Net	0.532	0.465	0.757	0.599	0.023	396.53	322.89
	ESPNet	<u>0.832</u>	<u>0.761</u>	<u>0.902</u>	0.899	0.031	98.62	<u>40.04</u>
	No KAN	0.809	0.744	0.894	0.860	<u>0.011</u>	94.50	66.10
	KAN-IMM only	0.800	0.739	0.889	0.840	<u>0.011</u>	90.70	68.20
	KAN-BAM only	0.804	0.740	0.892	0.854	<u>0.011</u>	<u>87.70</u>	60.70
	KAN-AINet (Ours)	0.859	0.795	0.919	<u>0.896</u>	0.007	48.40	29.10
PolypGen-C6 (unseen)	nnU-Net	0.637	0.552	0.805	0.725	0.050	419.75	241.59
	ESPNet	0.813	0.760	0.905	0.899	0.012	336.64	218.55
	No KAN	0.837	0.787	0.903	0.869	<u>0.014</u>	<u>89.90</u>	65.50
	KAN-IMM only	<u>0.856</u>	<u>0.806</u>	<u>0.906</u>	0.873	0.019	108.30	<u>41.80</u>
	KAN-BAM only	0.839	0.790	0.904	0.863	0.016	135.40	71.60
	KAN-AINet (Ours)	0.872	0.822	0.914	<u>0.892</u>	0.015	87.60	26.10
<i>Average across all unseen datasets</i>								
	nnU-Net	0.589	0.499	0.767	0.715	0.044	252.78	164.14
	ESPNet	<u>0.820</u>	<u>0.753</u>	0.898	0.886	0.025	160.94	93.43
	No KAN	0.792	0.729	0.876	0.829	<u>0.023</u>	101.63	73.60
	KAN-IMM only	0.810	0.750	<u>0.884</u>	0.837	<u>0.023</u>	<u>93.97</u>	<u>55.57</u>
	KAN-BAM only	0.797	0.735	0.881	0.833	<u>0.023</u>	109.37	70.03
	KAN-AINet (Ours)	0.838	0.776	0.898	<u>0.866</u>	0.020	71.43	35.73